

5.1.2 Neural Networks (NN)

نیورل نیٹ ورک ایک کمپیوٹر ماڈل ہے جو ڈیٹا سے پیٹرن سیکھتا ہے۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 03 of 15

★ Golden

5.1.2 Neural Networks (NN) ★ Golden

LEARN IT

A **neural network** is a computer model that **learns patterns from data**. It is **directly inspired by the human brain**, but it works as a **mathematical computer system** — not as living tissue.

Your brain has billions of nerve cells (**neurons**) that pass signals to each other. A neural network copies this idea with small processing units also called **neurons** or **nodes**. These nodes are **connected** and **pass information from one layer to another**, performing calculations at each step.

A typical neural network has three kinds of layer:

- **Input layer** — receives the raw data (pixels of a face, samples of a sound, marks of a student).
- **Hidden layer(s)** — the “thinking” in the middle. Nodes here combine the inputs and learn features (eye distance, nose shape, an edge in a photo).
- **Output layer** — gives the final answer (unlock / keep locked; “this is a cat”; “pass / fail”).

Input layer



Hidden layer(s)



Output layer

WORKED EXAMPLE

Practical example from the source: face unlock.

Step 1 — Input layer. You look at your phone. The camera captures your face as numbers (brightness of each pixel). Those numbers enter the input layer.

Step 2 — Hidden layers. The system studies face features such as **eye distance** and **nose shape**. Early hidden nodes may detect edges and shadows; later hidden nodes combine those into “this looks like Sara’s face.” It compares them with stored data from when you first set up the phone.

Step 3 — Output layer. If the features match, the phone **unlocks**. If they do not (your brother holds the phone), it stays locked.

The same three-layer story works for an NADRA-style biometric gate at a college: camera (input) → feature matching (hidden) → open the turnstile or not (output).

REMEMBER IT

I-H-O. Input receives, Hidden thinks, Output decides. “In Hidden Out.” A neural network is a **brain-inspired maths machine**, not a copy of a living brain.

COMMON MISTAKE

Mistake: “A neural network is an actual human brain inside the computer.”

Fix: It is only **inspired by** the brain. It is mathematics: numbers, multiplications and additions running on ordinary hardware.

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EXAM TIP

Very common questions: "What is a neural network?" (2–3 marks), "Name its three layers and explain each with the face-unlock example" (5 marks), and "How is a neural network related to the human brain?"

Full-mark recipe: (1) definition — computer model that learns patterns from data, inspired by the brain, mathematical; (2) neurons/nodes connected across layers; (3) Input / Hidden / Output, each in one sentence; (4) walk the camera → features (eye distance, nose shape) → unlock path. Draw the three boxes if the paper asks for a diagram.

CHECK YOURSELF

1. What are the small processing units of a neural network called?
2. In face unlock, which layer studies eye distance and nose shape?
3. Is a neural network a living brain? Explain in one sentence.

Answers: 1. Neurons or nodes. 2. Hidden layers. 3. No — it is inspired by the brain but works as a mathematical computer system.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A neural network is inspired by:

- a. The brain's networks of neurons (loosely)
- b. OSI cables
- c. Access forms
- d. Bubble sort

MCQ 2. It learns:

- a. Weights that turn inputs into outputs
- b. SQL indexes
- c. TCP windows
- d. Figma auto-layout

MCQ 3. Simple NNs have:

- a. Input, hidden, output layers
- b. Only a pie
- c. Only one Excel cell
- d. No numbers

MCQ 4. This is ★: define, then:

- a. One diagram + two uses
- b. Only careers
- c. Only MVPs
- d. Only keys

MCQ 5. NN vs linear formula:

- a. NN can stack non-linear layers for complex patterns
- b. NN cannot use data
- c. NN is a queue
- d. NN is HCI only

MCQ 6. Digits on paper are a classic:

- a. NN / vision example
- b. Primary key example
- c. Waterfall example
- d. K-map example

Extra short questions

1. Define neural network.
2. Three layer types.
3. Two applications.
4. Need for data.

Extra long question

1. Draw a 3-layer net for pass/fail from 3 marks. Label input, hidden, output. Explain one training idea in words (weights change). ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. A computer model of connected units that learns patterns from data, loosely like neurons.
2. Input, hidden, output.
3. Image recognition; speech; translation.
4. Without enough varied examples the net does not generalise.

Long

1. 3 input nodes, few hidden, 1 output. Training: compare prediction to label, adjust weights. Limit: not a biological brain.